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Wastewater reclamation for the production of construction bricks: technical, economic, and environmental feasibility

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ABSTRACT

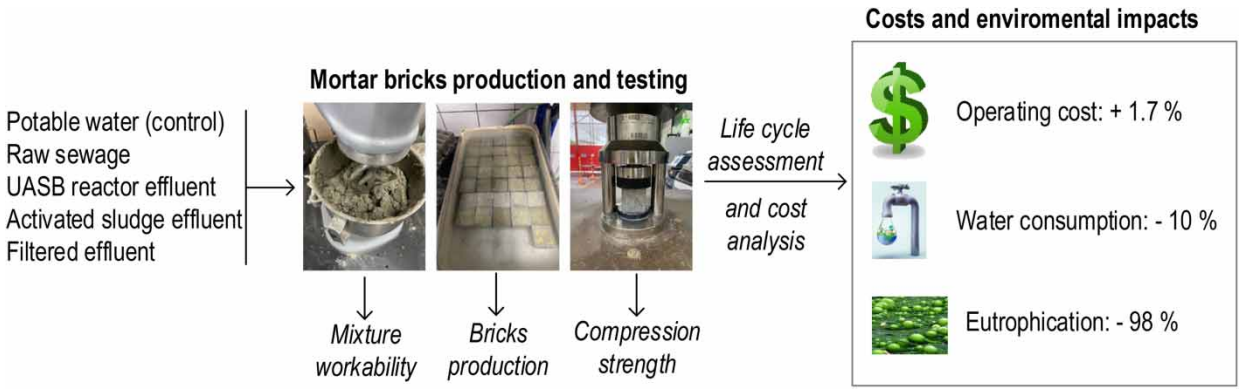
This study evaluates the technical, economic, and environmental impacts of using treated wastewater as a substitute for potable water (PW) in mortar bricks production. The study experimentally compared the reuse of raw sewage, UASB reactor effluent, activated sludge effluent, and filtered effluent to produce mortar bricks which were tested for mixture workability and compressive strength over curing periods of 3, 7, and 28 days. Slump values of the mixtures were close to 110 mm for all samples, and 28-day compressive strengths varied between 31.2 and 34.8 MPa, higher than the 17.2 MPa required for Type M mortars. Mortar made with activated sludge and filtered effluent exhibited properties comparable to those made with PW. Economic analysis revealed a slight cost increase of 1.7% due to effluent disinfection, but significant environmental benefits, such as reduced eutrophication due to the wastewater reclamation and water conservation, were noted, primarily due to the avoidance of discharging treated effluent into surface water bodies. These findings underscore the feasibility of using treated wastewater in construction, highlighting its potential to enhance sustainability in this field, and suggesting the necessity for further studies on the long-term effect of using reclaimed wastewater cementitious products.

Key words: circular economy, costs and life cycle assessment, mortar bricks, municipal wastewater reclamation, sustainable construction

HIGHLIGHTS

- Raw sewage and anaerobic effluent negatively affected the bricks' compressive strength.
- Activated sludge and filtered effluents maintained mortar quality.
- Slightly higher operating costs observed with treated effluent reuse compared with potable water.
- Noted significant environmental benefits, such as reduced eutrophication and water conservation.

GRAPHICAL ABSTRACT



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INTRODUCTION

Freshwater scarcity is a pressing global issue, particularly in arid regions, where four billion people face severe water shortages (Mekonnen & Hoekstra 2024). The construction industry, as a major consumer of freshwater, accounting for 16% of global water use, contributes significantly to this stress (Heravi & Abdolvand 2019). To mitigate water demand, alternative water sources, such as treated wastewater, are increasingly being considered.

Using treated effluent from sewage treatment plants (STPs) in concrete and mortar production could significantly reduce freshwater consumption in construction. However, concerns remain about the effect of reclaimed wastewater on the quality of mixtures and the final strength of the cast material. While treated wastewater is a viable alternative for construction, it may contain residual toxic substances, such as solids, nutrients, and organic pollutants, which could pose risks to both the environment and the structural integrity of concrete or mortar. These substances are typically reduced or removed during sewage treatment processes, though some may remain in the treated effluent (Metcalf & Eddy 2014). Importantly, many of these toxic substances can be immobilised within the mortar matrix, thereby reducing the likelihood of their release into the environment. Understanding the potential impact of these substances is crucial when considering wastewater for construction applications.

Previous studies have shown that, while wastewater may contain higher levels of impurities, the cast elements often fall within acceptable limits for mortar and concrete products, achieving comparable strength and durability to those made with potable water (PW) (Al-Jabri *et al.* 2011; Al-Joulani 2015). However, the effect of the level of sewage treatment on the quality of cementitious elements has been scarcely evaluated to date, raising questions about the engineering processes and associated costs required to ensure the quality of the final product. This issue can be significant for developing countries, where wastewater treatment and management pose major challenges.

In addition, over the last decade, the adoption of a circular economy in the sanitation industry has been proposed to enhance sustainable sanitation in developing countries by recovering valuable byproducts from sewage treatment, such as water, nutrients, and energy (Soares *et al.* 2019; AlSayed *et al.* 2020). Unlike the traditional linear model, where treated wastewater is often discharged into surface water bodies, potentially harming ecosystems (Anh *et al.* 2023), a circular approach promotes wastewater reclamation and valorisation. This not only prevents pollution but also provides economic and environmental benefits to local communities.

In this regard, this study evaluates the technical, economic, and environmental impacts of using effluent from STPs to produce mortar bricks. It was experimentally tested the workability and compressive strength of mortar bricks made with wastewater treated at different levels and conducted a cost analysis and life cycle assessment (LCA) to assess the potential benefits of this practice.

METHODS

Characterisation of the sampled wastewater

Three independent campaigns of sewage sampling were conducted for this study. For each campaign, five samples of water/municipal wastewater were considered: tap or PW (control), raw sewage (RS), anaerobic effluent (AE) (from an upflow anaerobic sludge blanket, UASB, reactor), activated sludge effluent (ASE) and filtered (1.5 μm pore size filter) ASE. These samples were taken from a municipal sewage treatment plant (STP) with an average flow of 250 m^3/d, located in San José, Costa Rica. The existing process in that STP consists of a preliminary treatment (4 mm sieve with a degritting channel), a UASB reactor, and an activated sludge post-treatment system. RS, AE, and ASE were directly sampled from this STP, whereas the filtered effluent was produced in the laboratory by filtering the ASE. In this study, only the activated sludge system effluent was subjected to filtration to simulate the tertiary treatment typically required in wastewater treatment plants (WWTPs). This approach reflects standard operational practices where aerobic post-treatment follows anaerobic processes, ensuring that the effluent meets discharge limits before additional treatment steps, such as filtration, are applied.

Table 1 shows the average characteristics of the wastewater samples used in this study.

Table 1 demonstrates that the wastewater samples used in this experiment underwent different levels of treatment, resulting in varying concentrations of organic matter. These characteristics align with the expected properties of these effluents, as described in the literature (von Sperling 2014). It is noteworthy that the ASE and the filtered effluent exhibited very similar characteristics, which can be attributed to the low concentration of solids in the ASE.

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Table 1 | Characteristics of the wastewater before its disinfection

	Wastewater sample (average value ± standard deviation)				
Parameter	**Potable water**	**Raw sewage**	**Anaerobic effluent**	**Activated sludge effluent**	**Filtered activated sludge effluent**
COD (mg/L)	6.5 ± 3.7	464 ± 33.2	241 ± 23.3	36 ± 2.8	40.5 ± 17.7
TSS (mg/L)	ND	164 ± 27	66 ± 17	13 ± 23	ND
pH (–)	7.48 ± 0.005	7.56 ± 0.27	7.18 ± 0.12	6.79 ± 0.34	6.90 ± 0.05
Conductivity (μS/cm)	129 ± 19	805 ± 167	843 ± 33	610 ± 36	514 ± 32

COD, chemical oxygen demand; TSS, total suspended solids; ND, not detected.

Although statistical analysis (Kruskal–Wallis tests) did not reveal significant differences between the effluent samples at the 5% significance level, this result is likely due to the small sample size (only three wastewater samples, one for each sampling campaign), which was constrained by the scope of this study. Despite the statistical findings, the practical quality differences among the effluents are pronounced and are supported by the literature (von Sperling 2014).

To fulfil local regulations, each wastewater sample was disinfected with a commercial sodium hypochlorite solution (4–5% concentration), adding the reagent until a residual chlorine concentration of approximately 1.0 mg/L was achieved. These disinfected samples were then used for the mortar brick production.

Production of the mortar bricks

After collecting the wastewater samples as described in the previous section, mortar bricks were produced in three independent campaigns. The same sand and cement were consistently used across all three campaigns and for all water/wastewater samples. The fine aggregate used was river sand, which was sieved using a No. 20 fine screen (0.85 mm) to obtain a finer and more homogeneous material, with a fineness modulus of 2.9. The specific gravity of the sand, determined following the ASTM C128 test, was 2.50. Additionally, the aggregate absorption was measured using the ASTM C128 procedure, yielding a value of 3.58%. These values are typical for fine aggregate, and the absorption value was used to determine the volume of water required for mixture preparation.

The cement used in this experiment was Portland cement, commercially available in Costa Rica under the brand name *Cemento Fuerte de Holcim*. This specific cement was chosen for its properties suitable for mortar preparation. The relative density of the cement, measured according to ASTM C188, was 2.77, closely aligning with the manufacturer's specification of 2.75–3.00. The initial and final setting times, determined following ASTM C191 (with a water-to-cement ratio of 0.3, with PW), were 160 and 285 min, respectively. These values fall within the recommended range for the setting time of hydraulic cement by the Vicat needle test, which specifies an initial setting time of over 45 min and a final setting time of under 6 h, ensuring the mixture's plasticity during application and its initial gain of rigidity.

The mixture design followed ASTM C109 (Standard Test Method for Compressive Strength of Hydraulic Cement Mortars). The mortar was composed of 1 part cement to 2.75 parts sand by mass, with a cement-to-water ratio of approximately 0.485 by mass. This ratio allowed for a slump value (workability indicator) of 110 ± 5 mm, as determined in preliminary tests using PW, and in accordance with ASTM C109.

Each of the three campaigns was an identical repetition (only the wastewater sample was different, from a new sampling campaign). A total of 180 mortar cubes were produced in this study, with 60 cubes per experimental campaign. For each campaign, the cubes were equally distributed among the five different water categories: PW (control), RS, UASB reactor effluent, ASE, and filtered effluent, resulting in 12 cubes per category per campaign. The mortar mixtures were characterised in terms of workability by measuring the slump value according to ASTM C1437. The bricks, generated with the five different water qualities (potable, RS, AE, ASE, and filtered ASE), were tested for compressive strength in accordance with ASTM C109 at 7, 14, and 28 days. Figure 1 shows photographs of the mixture production and brick failure testing.

As shown in Figure 1, bricks produced in this study were moulded into 50 mm (2-inch) cubes, as specified by ASTM C109 for compressive strength testing of hydraulic cement mortars. The cubes, regardless of the water/wastewater used, exhibited a uniform grey colour typical of mortar, with a smooth and consistent texture across all samples. After moulding, the mortar

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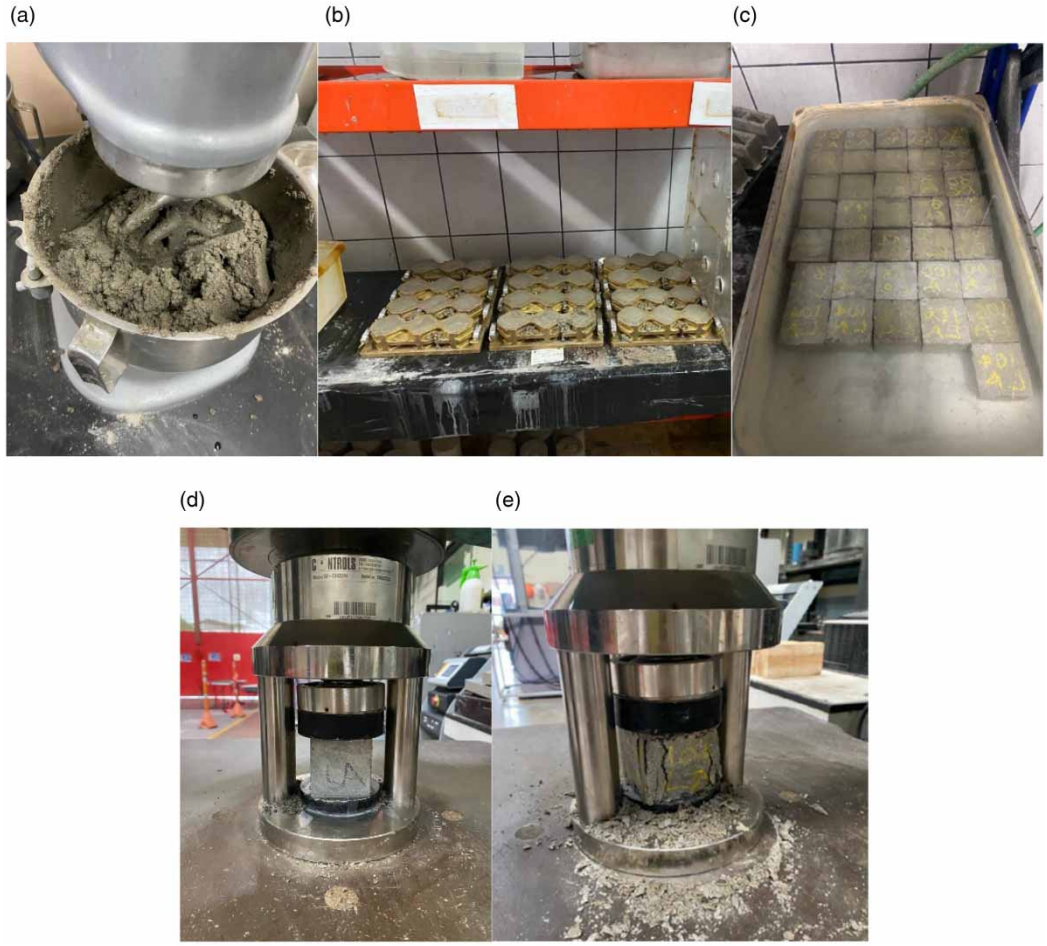


Figure 1 | Experimental set-up of the study: (a) mixture preparation; (b) brick moulding; (c) finished bricks; (d) brick before compression strength test; and (e) brick after compression strength test.

cubes were placed in a moist room at 22 $\pm$ 0.5 °C and 95% relative humidity, where they were maintained under controlled temperature and humidity conditions throughout the entire curing period. This process ensured optimal hydration and curing, preventing any premature drying or cracking.

Experimental data analysis

The experimental results were statistically analysed using non-parametric statistics, employing the software Statistica v.10 (STATSOFT 2011). Kruskal–Wallis statistical tests with multiple comparisons of the medians of slump value and compressive strength for each water/wastewater quality were applied. A confidence level (α) of 95% was considered for the statistical tests. Additionally, the results from these tests were compared against the classification of mortar types specified in ASTM C270, which categorises mortars into different types based on their compressive strength at 28 days. These types include Type M, S, N, O, and K, with Type M being the strongest, requiring a minimum compressive strength of 17.2 MPa.

Considerations for the economic and environmental analysis

Figure 2 presents a diagram of the scenarios considered in this analysis.

Two independent scenarios were evaluated, with the functional unit being 1 ton of mortar bricks. In Scenario 0, PW was used to produce 1 ton of bricks, and non-disinfected treated effluent was discharged into a stream or river. This conventional approach consumes clean water resources. In Scenario 1, disinfected treated wastewater, using a 15% sodium hypochlorite solution, replaced PW in the brick production process. This scenario aims to reduce clean water consumption during brick production. In both scenarios, sand, cement, and energy were considered for brick production.

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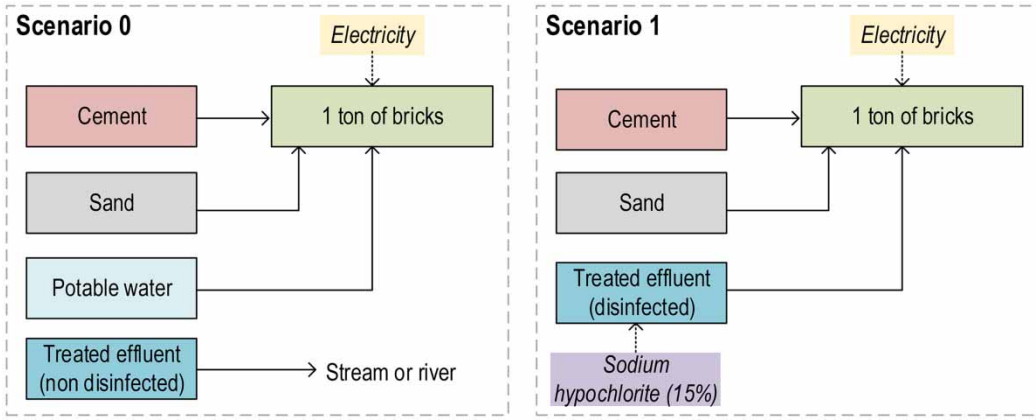


Figure 2 | Scenarios for the economic and environmental analysis considered in this study: (a) Scenario 0: use of potable water for the brick generation and (b) Scenario 1: treated wastewater reclamation for the brick generation.

Table 2 shows the inventory considered for the economic and environmental analysis.

For the LCA analysis, the SIMAPRO software (PRé Sustainability 2024) version 9.4.0.2 was used. The database considered for the inventory was Ecoinvent 3 (Ecoinvent 2024). When specific Costa Rican parameters were available in Ecoinvent, they were used; otherwise, 'Rest of World' parameters were applied. The PW and electricity were assumed to be tap water {RoW} market, and low voltage electricity {CR} (Costa Rica) market for electricity in the Ecoinvent database, respectively. For sand and cement, the Sand {RoW} market and cement Portland {RoW} market were considered, respectively. Sodium hypochlorite in a 15% solution state {RoW} was used as the disinfection reagent (Scenario 1). For Scenario 0, the discharge of the treated effluent (emission to water) that would have been reclaimed (Scenario 1) was added, considering a volume of water equal to the non-reclaimed flow (0.115 ton), with mass flows for the quality parameters of BOD, COD, total suspended solids, total nitrogen (TN), and total phosphorus (TP) of 2.06, 4.12, 1.49, 3.44, and 0.773 g, respectively. These values were calculated considering the water volume required to produce 1 ton of mortar bricks, and the experimental average concentrations of the ASE (Table 1), with typical values available in the literature (von Sperling 2014) for this technology of BOD (half of COD), TN (30 mg/L), and TP (5 mg/L).

For the evaluation of environmental impacts, the 'ReCiPe 2016' model (Huijbregts *et al.* 2017) was considered, with a midpoint type and a hierarchical perspective, due to its widespread use in LCA studies with WWTPs and adherence to the recommendations of the literature in the field (Corominas *et al.* 2020). This model allows for the calculation of a total of 18 impacts based on the inventory data: ecosystem damage ozone formation, fossil resource scarcity, freshwater ecotoxicity,

Table 2 | Inventory for the economic and environmental analysis

Parameters	Considered values	Comments	Reference
Inflows to produce 1 ton of bricks	Cement: 0.236 ton Sand: 0.649 ton Water/wastewater: 0.115 ton Energy: 1.85 kWh NaClO volume (15%): 1.03 L	W/C ratio of 0.485; 2.75 parts of sand per cement in mass; electric gear of 5 HP working during 30 min	Experimental results of this study
Costs	Cement: 240.5 US $/ton Sand: 104.2 US$/ton Potable water: 3.33 US$/ton Electricity: 0.22 US $/kWh	Local costs in Costa Rica	EPA (2023) ARESEP (2024)

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freshwater eutrophication, global warming, human carcinogenic toxicity (HCT), human damage ozone formation, human non-carcinogenic toxicity, ionising radiation, land occupation, marine ecotoxicity, marine eutrophication, mineral resource scarcity, particulate matter formation, stratospheric ozone depletion, terrestrial acidification, terrestrial ecotoxicity, and water consumption. Normalised impacts were also calculated to compare the damages between different impacts, using the normalisation factors specific to ReCiPe 2016.

RESULTS

Experimental results of workability and strength of the casted elements

Figure 3 illustrates the slump values of concrete mixtures made with different qualities of the treated sewage, reflecting the workability of the three samples.

Average slump values of 110.1 ± 3.1 mm, 109.6 ± 3.2 mm, 109.4 ± 3.2 mm, 110.3 ± 3.1 mm, and 111.0 ± 2.6 mm were obtained for PW, RS, AE, ASE, and FE, respectively. Slump values of 110 ± 5 mm are recommended to guarantee easier handling and placement in construction of the cementitious mixture (ASTM C109). In this regard, for the water-to-cement ratio of 0.485, all the wastewaters demonstrated adequate workability, and no statistically significant differences between the median slump values were observed at a 95% confidence level (p-value of 0.4633).

Figure 4 presents the compressive strength results of the mortar bricks cast with the different types of wastewaters over three curing periods: 3, 7, and 28 days, substituting 100% of the PW with the different STP effluents. The results are shown for three series of tests, with an additional chart summarising the overall results. Each bar chart provides the average compressive strength values for the samples, allowing for a comprehensive comparison across the different wastewater types.

As shown in Figure 4, for the overall results, at the 3-day curing period, the bricks made with PW exhibited the highest compressive strength, reaching a mean of 15.3 ± 1.1 MPa. The cubes made with RS showed a lower strength of 12.2 ± 1.5 MPa, indicating a 20% reduction. The AE bricks achieved a compressive strength of about 12.6 ± 1.5 MPa, showing an 18% reduction compared with PW. The ASE and activated sludge filtered effluent (FE) bricks exhibited average strengths of 13.4 ± 1.6 MPa and 13.9 ± 1.9 MPa, respectively, with the FE cubes showing a 17% reduction compared with the PW cubes. When the median values of each sample were compared for the overall results, statistically significant differences at a 95% confidence level were observed (p-value of 0.0004). Statistically significant differences were found only between PW and RS (p-value of 0.000692) and between PW and AE (p-value of 0.002801). This initial data suggests that while RS and AE weaken the cementitious mix, ASE and FE offer an acceptable performance, which is still lower than that obtained with PW, although not statistically different.

At the 7-day curing period, all samples displayed increased compressive strength. The PW bricks reached an average compressive strength of 21.4 ± 1.7 MPa. The RS bricks improved to 19.4 ± 1.0 MPa, narrowing the gap but still showing a 9.3% reduction. The AE bricks exhibited a strength of 19.1 ± 1.4 MPa, while ASE and FE cubes reached about 19.3 ± 1.8 MPa and 19.8 ± 1.8 MPa, respectively. The statistical tests showed the same pattern as obtained for the 3-day curing period, with statistically different medians (p-value of 0.0150) and significant differences only between PW and RS (p-value of 0.04037) and between PW and AE (p-value of 0.0348).

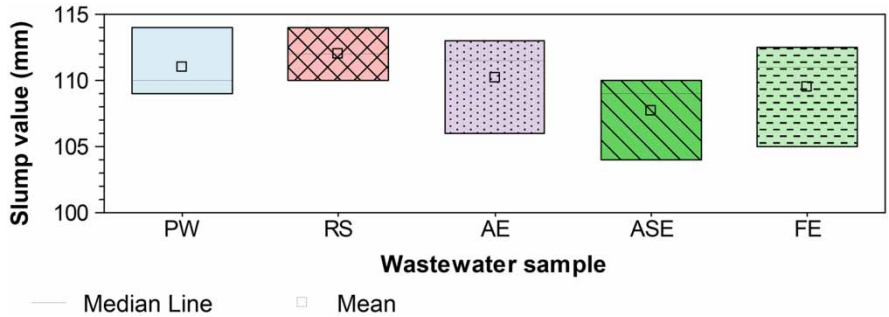


Figure 3 | Slump values to assess the mixture workability for the different wastewater samples. PW, potable water; RS, raw sewage; AE, anaerobic effluent; ASE, activated sludge effluent; FE, activated sludge filtered effluent.

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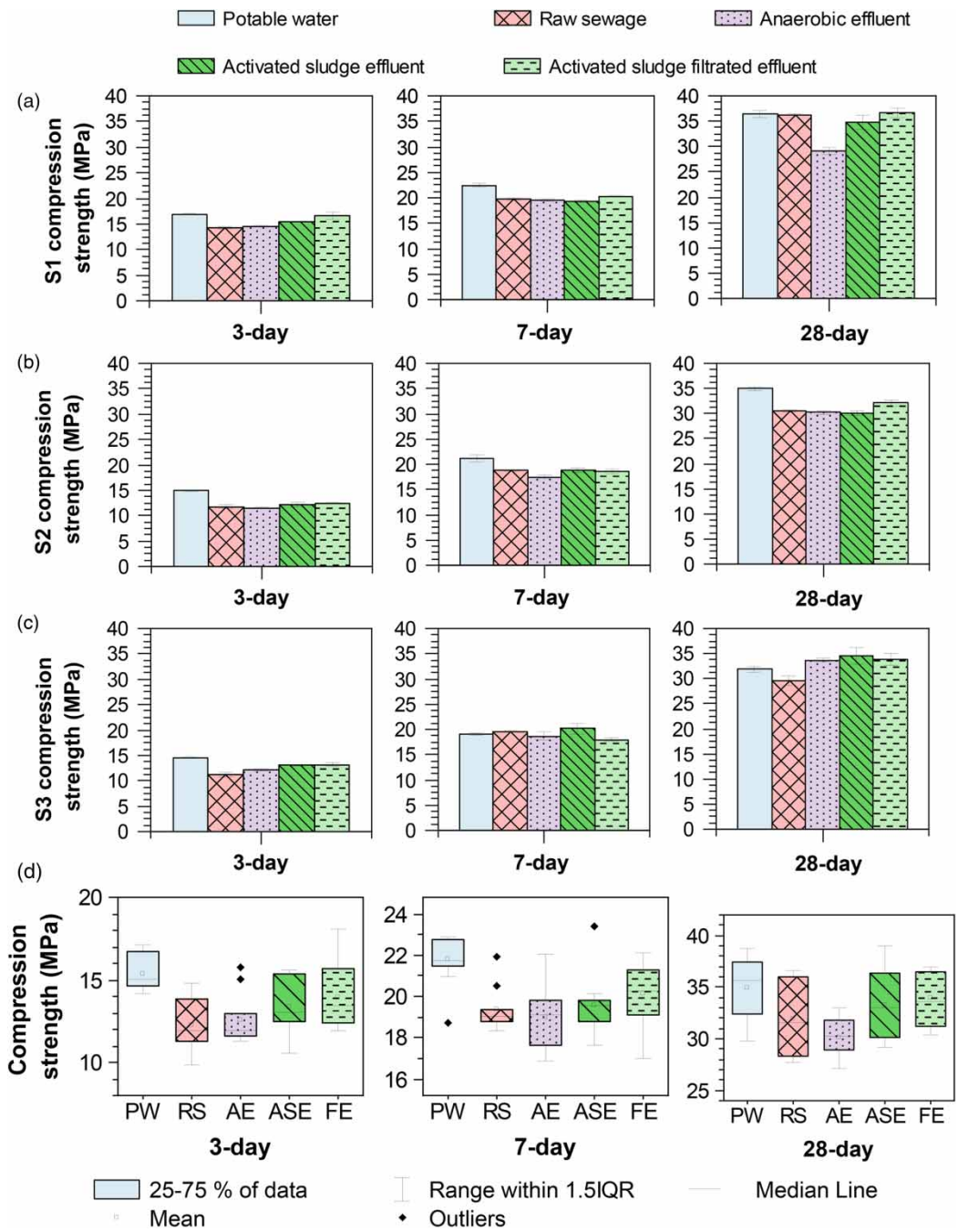


Figure 4 | Compression strength of the casted elements: (a) first series; (b) second series; (c) third series; (d) overall results. PW, potable water; RS, raw sewage; AE, anaerobic effluent; ASE, activated sludge effluent; FE, activated sludge filtered effluent.

By the 28-day curing period, all samples achieved their maximum compressive strength. The PW bricks showed the highest strength at approximately 34.8 $\pm$ 2.8 MPa. The RS bricks reached about 31.8 $\pm$ 3.2 MPa, reflecting an 8.6% reduction. The AE bricks displayed a strength of approximately 31.2 $\pm$ 2.4 MPa, while the ASE and FE bricks achieved strengths of 33.3 $\pm$ 3.6 MPa and 33.8 $\pm$ 2.4 MPa, respectively. The statistical tests showed differences between the medians (p-value of 0.0094), with significant differences at the 95% confidence only between PW and AE, indicating that no significant differences were observed between PW and RS (p-value of 0.251). However, it should be noted that a compressive strength above 21 MPa at 28 days was obtained for every wastewater type, which meets the minimum requirement for structural

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concrete (ACI Committee 318 2019) and for Type M mortar according to ASTM C270, which requires at least 17.2 MPa. A comparison to other similar studies is presented in the Discussion section.

The analysis of these results underscores the viability of using treated wastewater, particularly ASE and FE, in mortar production, as no statistically significant differences (at a 95% confidence level) were observed between these samples and the samples produced with PW. This is especially relevant in regions facing water scarcity, where the conservation of PW is crucial.

Conversely, the use of RS and AE would reduce the concrete resistance, arguably due to the presence of organic matter that affects the casting process. This could compromise the structural integrity when using these low-quality effluents for structural elements. This is further examined in the Discussion section.

Economic and environmental considerations

For the analysis of economic and environmental impacts, only the scenarios involving PW (Scenario 0) and effluent from the activated sludge system (Scenario 1) were considered for comparison. This decision was made to simplify the analysis by focusing on an effluent that provided an optimal balance between treatment cost and quality. The ASE was selected because it is less costly than filtration, achieved a quality equivalent to PW, and including filtered effluent would have introduced redundancy in the analysis due to the similar quality and results. Detailed information on these models, such as the inventory and unitary costs, is provided in the Materials and Methods section. Table 3 summarises the key results of the cost comparison.

According to Table 3, despite the introduction of treated effluent in Scenario 1, the overall cost remains slightly higher than for Scenario 0. The minor increase in total cost (US$ 2.13, only 1.7%) can be attributed primarily to the disinfection reagent required to treat the effluent, which would not be necessary if the final effluent were discharged into the receiving water body. However, the costs of cement, sand, and electricity remain constant across both scenarios, indicating that the primary materials and energy requirements are unaffected by the water source used. The higher cost could be offset by the environmental benefits, as explained in the Discussion section.

Figure 5 and Table 4 show the relative and absolute global environmental impacts reported values for Scenarios 0 and 1, respectively.

Figure 5 illustrates that the dominant contributions to the environmental impacts in both scenarios are from cement and sand, regardless of the water source. These materials significantly affect all impact categories except for freshwater and marine water eutrophication. In Scenario 0 (PW used), the treated effluent discharge into the environment has the most substantial impact on these eutrophication categories. The use of chlorine, which is specific to Scenario 1 (reclaimed effluent), and the electricity consumption contribute to certain impact categories, but remain relatively minor compared with the other activities.

This trend is confirmed by Table 4, which shows minimal differences between the two scenarios across most impact categories. For instance, the global warming potential shows a slight reduction of 0.12% when reclaimed effluent is used instead of PW. Similar minor differences are observed for stratospheric ozone depletion (−1.14%), ionising radiation (−1.28%), and fine particulate matter formation (−0.32%). Other impact categories, such as terrestrial acidification (−0.23%), terrestrial

Table 3 | Cost comparison analysis for the elaboration of 1 ton of bricks

	Scenario 0: Potable water		Scenario 1: Reclaimed effluent	
	Cost (US$)	%	Cost (US$)	%
Potable/tap water	0.38	0.30	–	–
Disinfection reagent	–	–	2.51	1.97
Cement	56.80	45.4	56.80	44.60
Sand	67.64	54.0	67.64	53.11
Electricity	0.41	0.33	0.41	0.32
Total	**125.24**	**100.00**	**127.37**	**100.00**

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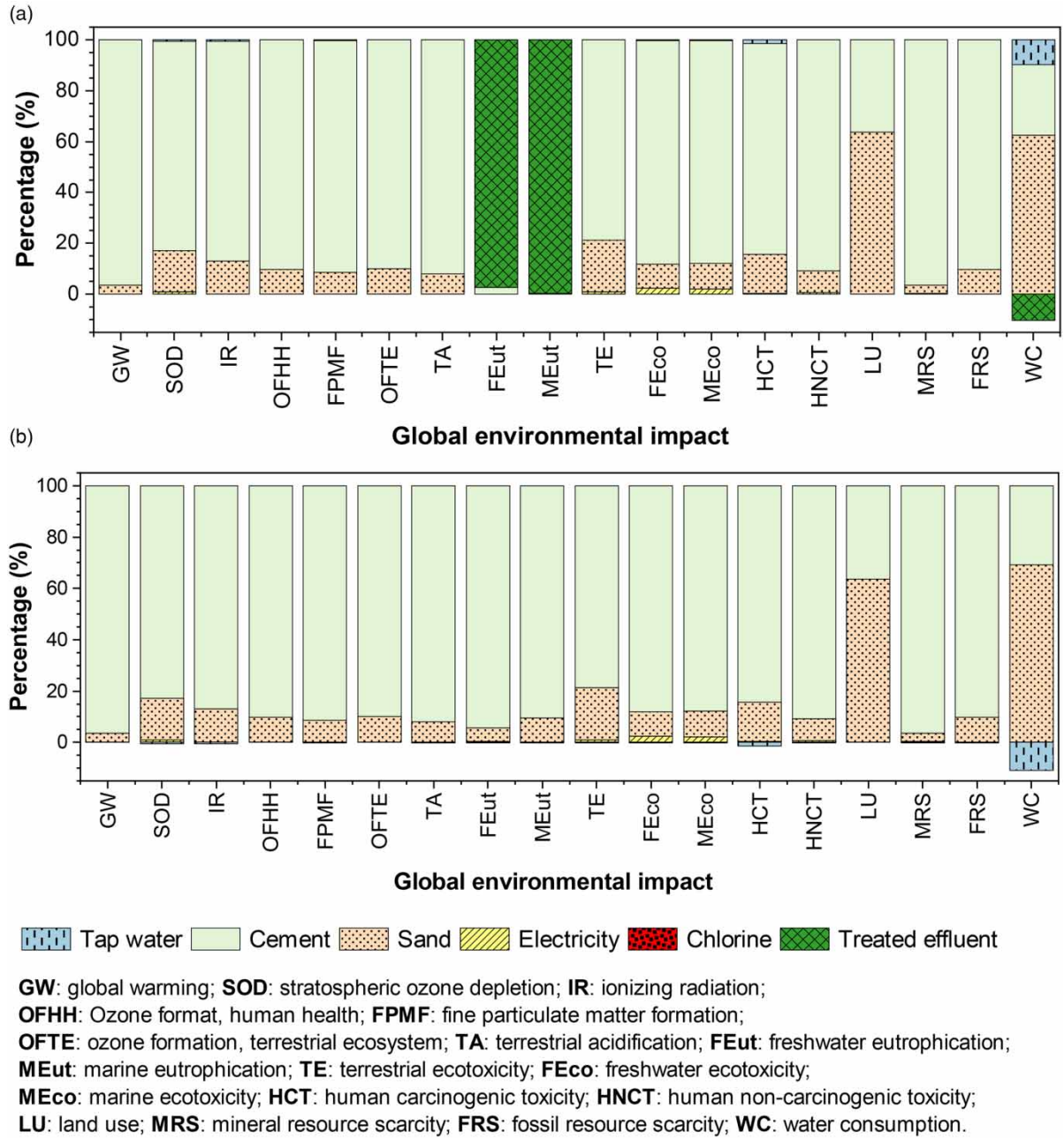


Figure 5 | Relative global environmental impacts for (a) Scenario 0 (potable water) and (b) Scenario 1 (reclaimed effluent), during the production of 1 ton of bricks.

ecotoxicity (−0.22%), and marine ecotoxicity (−0.37%), show marginal reductions, all below 1%. However, a significant reduction is seen in freshwater and marine eutrophication, where using reclaimed effluent decreases the impact by 97.3 and 99.8%, respectively, reflecting a substantial benefit of water reuse in terms of reducing nutrient load in freshwater and marine bodies. The impact on water consumption is reduced by 10.3%, highlighting another significant advantage of reusing effluent for conserving water resources.

Figure 6 shows the normalised global environmental impacts for Scenarios 0 and 1. It is noticeable that the most significant impacts are related to freshwater ecotoxicity (FEco), HCT, and marine ecotoxicity (MEco). The reuse of effluent in Scenario 1 significantly reduces freshwater and marine eutrophication impacts, reflecting a notable environmental benefit. For HCT and

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Table 4 | Reported values for the global environmental impacts of each scenario

Environmental impact	Units	Reported values		
		Scenario 0 (potable water)	Scenario 1 (reuse water)	Diff. (%)
Global warming	kg CO_2 eq	$2.33 \cdot 10^{2}$	$2.33 \cdot 10^{2}$	−0.12
Stratospheric ozone depletion	kg CFC11 eq	$1.68 \cdot 10^{-5}$	$1.66 \cdot 10^{-5}$	−1.14
Ionising radiation	kBq Co-60 eq	1.61	1.59	−1.28
Ozone formation, human health	kg NO_x eq	0.538	0.537	−0.14
Fine particulate matter formation	kg PM2.5 eq	0.179	0.178	−0.32
Ozone formation, terrestrial ecosystems	kg NO_x eq	0.548	0.547	−0.14
Terrestrial acidification	kg SO_2 eq	0.434	0.433	−0.23
Freshwater eutrophication	kg P eq	$8.40 \cdot 10^{-1}$	$2.24 \cdot 10^{-2}$	−97.3
Marine eutrophication	kg N eq	1.02	$2.17 \cdot 10^{-3}$	−99.8
Terrestrial ecotoxicity	kg 1.4-DCB	$3.42 \cdot 10^{2}$	$3.42 \cdot 10^{2}$	−0.22
Freshwater ecotoxicity	kg 1.4-DCB	2.73	2.72	−0.38
Marine ecotoxicity	kg 1.4-DCB	3.78	3.76	−0.37
Human carcinogenic toxicity	kg 1.4-DCB	4.07	3.95	−2.92
Human non-carcinogenic toxicity	kg 1.4-DCB	80.2	79.9	−0.31
Land use	m^2a crop eq	4.73	4.72	−0.10
Mineral resource scarcity	kg Cu eq	0.897	0.895	−0.21
Fossil resource scarcity	kg oil eq	25.6	25.5	−0.27
Water consumption	m^3	1.00	0.901	−10.3

other environmental impacts, both scenarios exhibit very similar behaviours due to the common use of cement and sand in brick production. Cement use continues to be the main contributor to these other impacts.

DISCUSSION

Table 1 in this study provides a characterisation of the wastewater samples used, including key parameters such as COD, TSS, pH, and conductivity. The variations in these parameters among the different samples are crucial to understanding their impact on the mechanical properties of the mortar bricks produced. Specifically, the RS and AE samples, which exhibited higher levels of COD and TSS, showed reduced compressive strength in the resulting mortar bricks. This reduction is attributed to the presence of higher organic and suspended solid content, which can interfere with the hydration process and, consequently, the development of strength in the cement matrix.

Previous studies (Ma *et al*. 2016; Rashid *et al*. 2018; Sakir *et al.* 2020) have stated that the presence of organic impurities in cement-based composite products (e.g., mortar bricks), such as proteins, humic and fulvic acids, among others, can lead to a reduction in compressive strength over time. This reduction is often attributed to the interference of organic materials with the hydration process of the cement, which is crucial for developing the mortar's mechanical properties. For instance, proteins and other organic substances can disrupt the formation of calcium silicate hydrate (C-S-H), the primary binding phase in cementitious materials, resulting in a less dense and weaker matrix that gains resistance more slowly. Moreover, the presence of organic acids can lead to chemical reactions that further deteriorate the structural integrity of the mortar. These acids can react with calcium hydroxide in the cement paste, forming calcium salts that lack the binding strength of the original compounds. This process, known as decalcification, results in a porous and weakened mortar structure, compromising its long-term durability (Rashid *et al*. 2018).

In contrast, the ASE and FE samples, with their lower COD and TSS levels, produced mortar bricks with compressive strength values closer to those obtained using PW. These findings suggest that the removal of a significant portion of organic and solid contaminants, as achieved through activated sludge treatment and subsequent filtration, can yield effluents that are more suitable for use in mortar production.

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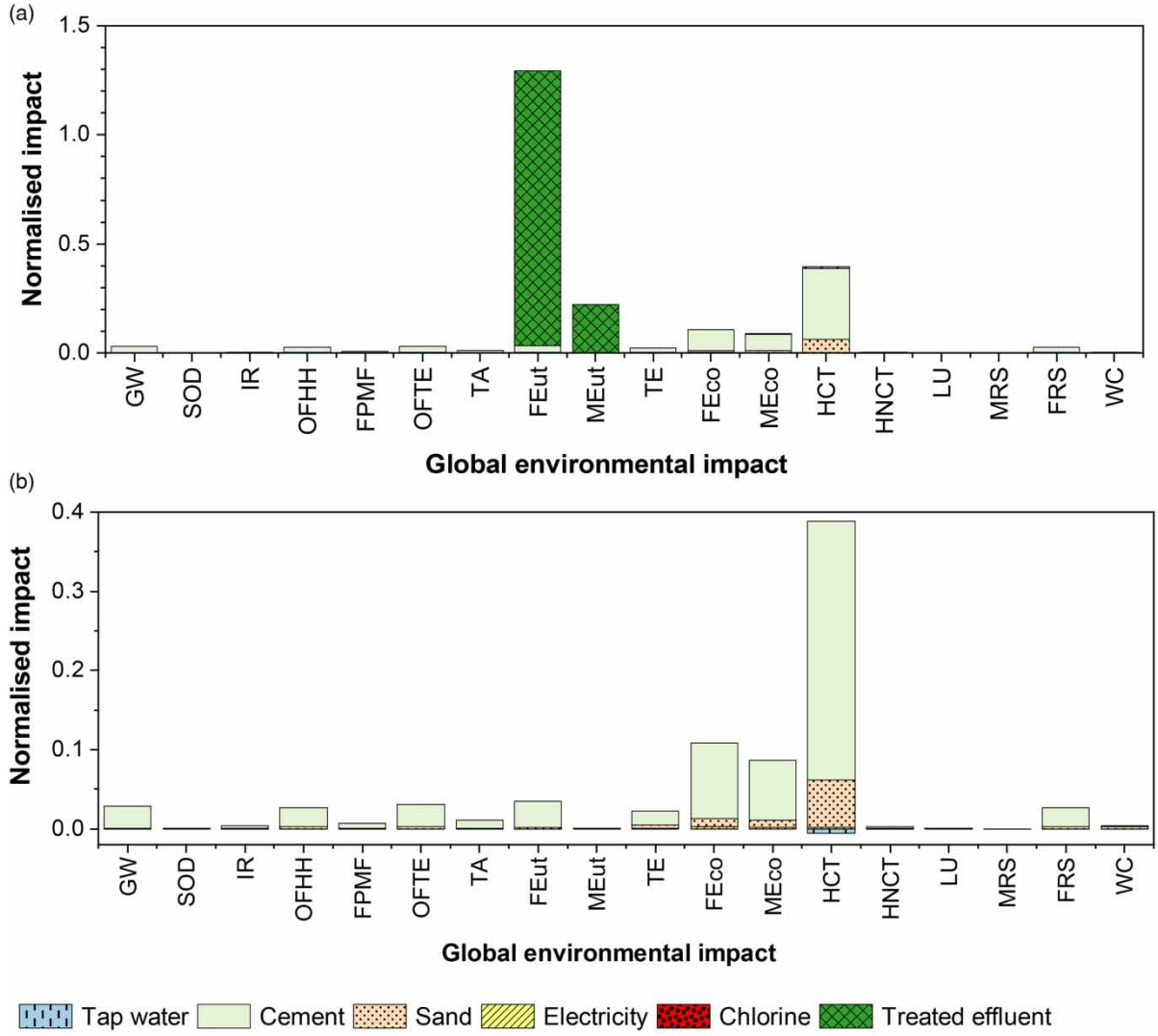


Figure 6 | Normalised global environmental impacts for (a) Scenario 0 (tap water) and (b) Scenario 1 (reclaimed effluent), during the production of 1 ton of bricks.

Thus, this study suggests that using effluent from the activated sludge system (ASE and FE) as a substitute for PW in mortar production would not negatively affect the final product. The experimental results demonstrated that the workability and compressive strength of concrete produced with ASE and FE are comparable to those made with PW. These findings are consistent with previous studies, such as Al-Jabri *et al.* (2011), Al-Joulani (2015), and Meena & Luhar (2019), which reported minimal impact on the concrete properties when using treated wastewater of that treatment level (secondary or tertiary). It should be noted that the mixture workability was not statistically different between the wastewaters of variable quality and the PW.

Similar studies, such as those by Ghrair & Al-Mashaqbeh (2016) and Rashid *et al.* (2018), have also demonstrated that the use of treated wastewater can result in mortar compressive strengths that meet or exceed conventional standards, further

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indicating the viability of reclaimed wastewater use in mortar production. In these studies, the compressive strengths at 28 days were also above 17.2 MPa (the minimum required for Type M mortar, as per ASTM C270) for both the control (PW) and treated effluent. The effect of different quality effluents was not evaluated in these studies, and slightly lower values were obtained for the wastewater samples.

Regarding the economic analysis and LCA, the obtained results indicated that wastewater reclamation would incur slightly higher costs (+1.7%). The higher economic cost of using treated wastewater instead of PW is primarily due to the low cost of tap water, which is significantly lower than the cost of the chlorine required for effluent disinfection. This disinfection is necessary to ensure the safety of construction operators and to comply with local regulations. For using reclaimed wastewater to be economically advantageous, the cost of PW would need to be as high as US$ 22.5 per cubic metre, which is approximately three times higher than the highest reported cost of tap water globally (Burgueño Salas 2021). While the cost difference between using treated wastewater and PW may be minor, especially considering that tap water is often subsidised, the reuse of treated wastewater presents a highly valuable and sustainable option, particularly in arid regions where water scarcity is a critical concern.

Despite the slightly higher cost, using the wastewater reclamation to produce bricks would have significant environmental benefits. This practice would notably reduce the eutrophication impacts on freshwater and marine bodies, as well as water consumption. The reductions in other environmental impacts are minimal, given that cement and sand remain the main contributors to these impacts. Previous studies have shown the environmental benefits of using waste such as silica fume with reclaimed water (Delnavaz *et al.* 2022) and recycled aggregates (Kurda *et al*. 2018). However, the effect of only substituting the water in the mixture has not been evaluated to date, to the best of our knowledge. In this case, the larger environmental benefit was derived from the avoidance of discharging treated wastewater, still containing nitrogen and phosphorus which contribute to eutrophication, into surface water bodies.

CONCLUSIONS

The study demonstrated the potential of using treated wastewater as an alternative to PW in mortar brick production, aligning with sustainable construction practices. This approach not only helps conserve freshwater resources but also provides a viable method for repurposing treated effluent. Technically, the use of treated sewage, especially from activated sludge systems, has shown comparable workability and compressive strength to that of concrete made with PW, indicating its feasibility for practical applications.

Economically, the initial costs may be slightly higher due to the need for disinfection, but the cost difference compared with PW is minor, especially considering that tap water is often subsidised. In this context, treated wastewater reuse remains a promising and valuable option, particularly in coastal areas or arid regions with water scarcity, where it can help meet the elevated demand for water.

In addition, from an environmental perspective, the use of reclaimed wastewater in construction offers substantial benefits. It significantly reduces the eutrophication impacts on freshwater and marine ecosystems, primarily by preventing the discharge of treated effluent into these water bodies. The practice also conserves water resources, which is crucial in regions facing water scarcity.

Future research should focus on optimising treatment processes and exploring the long-term performance of cementitious elements made with reclaimed wastewater under various environmental conditions. Such efforts can further solidify the role of wastewater reclamation in promoting environmentally sustainable construction methods.

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AUTHOR CONTRIBUTIONS

E.C. conceptualised the study, wrote and edited the article. C.M. conducted the experimental work. N.C. reviewed and edited the article.

DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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